PALEOMAGNETIC EVIDENCE FOR ROTATION OF NORTHERN CALABRIA

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Abstract. A mean paleomagnetic direction of D = 286,6 and I = 44,5 (α_{95} = 17°, pole position at 297 E, 16 N) was obtained from Upper Cretaceous limburgites at M.Cerviero, Northern Calabria. The amount of counterclockwise rotation of the area since the Upper Cretaceous is of about 73°, larger than any known rotation of the surrounding areas: Sicily, Apulia and Campania. According to this data Northern Calabria, which is considered to be wholly allochtonous on geologic grounds, was subject to further differential displacement in respect with Sicily which was relatively fixed to the African plate, as well as in respect with the rotations between the Apennines, the Sardinian-Corsican microplate and Africa.

Introduction

According to paleomagnetic data most crustal blocks of the central Mediterranean area underwent counterclockwise rotations in respect with stable Europe since Permian to Tertiary times (Zijderveld & Van der Voo, 1973). The pre-rotational pole positions calculated for Sardinia, Sicily, the Southern Alps, the Apulia platform and the Apennines have been relatively close to the African poles till the Tertiary, when the alignment with the European pole positions occurred (De Jong et al., 1973, Manzoni, 1975) possibly as a result of a general twist of the Mediterranean area. However the rotations between different areas seem to be differential (Channel & Tarling, 1975, Lowrie & Alvarez, 1975, Klootwijk C.T. & Van den Berg, 1975).

Within this picture of geodynamic motions, Northern Calabria clearly shows its geological allochtony since it is constituted by a complex pile of nappes of sedimentary, metamorphic and crystalline terrains. To some authors (Alvarez et al., 1974) the Calabrian crust once belonged to an Alpine belt which was fragmented and displaced further to the east and to the south of the Sardinian-Corsican microplate.

Geology and sampling

48 oriented hand samples from 10 sampling sites of basic effusive and intrusive rocks, "limburgites", were taken on M. Cerviero (lat. 39,9N and long. 16,02E) a massive of Mesozoic limestones and Tertiary sediments which belongs to the Verbicaro tectonic unit auct., comprised within a pile of nappes superposed to each other along overthrust surfaces, the Panormide Complex of Ogniben 1969.

The relationships of the tectonic units are described by Dietrich & Scandone, 1972. Descriptions of the limburgites, aggregates of serpentine minerals, pyroxenes, calcite, plagioclase and other minerals are found in Quitzow 1935 and Lucini et al. 1957. The age of the limburgites, debated in the past, was established by Cortini et al. 1974 as uppermost Cretaceous (Maastrichtian-Palaeocene), and the unit underwent a later metamorphic event, dated by the K-Ar method to 18.5±0.09 m.y., during which the rock was heated to over 300°C.

5 additional samples were collected in the overlying tectonic unit, 15 Km east of M. Cerviero at Colle Vaccaro (site 11 of table I).

Tests and results

Step demagnetization in AC fields up to 2,600 Gauss was carried out on 2 to 4 pilot specimens for each site. Fig. 1 shows three types of curves of intensity decay: type a (dashed line) with hard and extremely stable magnetization, type b (continuous line) with rapid decay of intensity and a minor component of hard remanence, type c (dotted line) with destruction of remanence within 200 Gauss. Curie temperature of hematite were clearly detected at sites 4 and 7 (type a curves) and of magnetite ai sites 8 and 9 (type b curves) by a magnetic balance. All 13 specimens were demagnetized in steps according to the magnetic properties, and optimum clusterings of directions were calculated. Site 8 was discarded because of unstable magnetization.

The directions are arranged into two sets; the major group, with directions corrected for dip and including the sites with high stability of magnetization represents the primary directions of the limburgites with westerly and north-westerly declinations. The significance and reliability of the data from the three sites with north and north-easterly directions meets some questions: low stability for site 6; locations close to fault and overthrust zones for sites 2 and 10. On these grounds data from sites 6, 2 and 10

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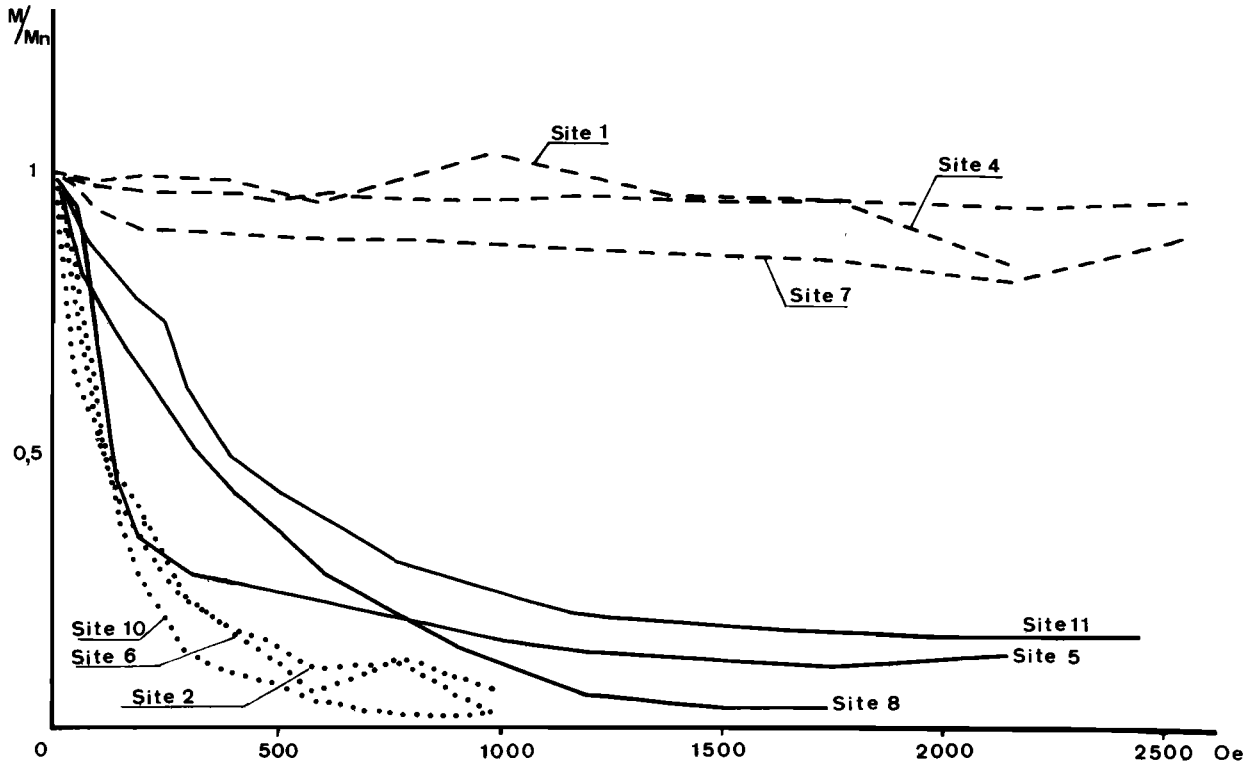


Fig. 1 - Decay of intensity of magnetization of basic rocks from Northern Calabria

is not included in the mean of selected sites. The direction of site 11 is again oriented to the west.

Interpretation

The paleomagnetic directions from the limburgites of Northern Calabria are displaced further to the west than those of comparable age from the Umbrian Apennines (344,2, + 40,5), Campania (322,4, + 28,2), Apulia (342,4, + 36), Eastern Sicily (165,4, - 24,3), Southern Prealps (343, + + 41,4), and of Tertiary from Sardinia (322,1, + 46,9). This fact implies that the unit including the limburgites rotated further counterclockwise since the Maastrichtian than the other known Italian areas did in respect with stable Europe.

The total amount of rotation, between Upper Cretaceous times and today's field direction, is quite close to that calculated in the reconstruction shown by Alva-

Table I - Paleomagnetic data from M.Cerviero after AC cleaning

S	N	n	Oe	M e.m.u. x 10^{-6}	D	I	α_{95}	C
1	2	5	400	580	289	+ 56,8	-	x
2	6	11	250	19,4	20,9	+ 60,8	12°	
3	5	16	600	188	252,3	+ 50,9	51°	x
4	5	10	1600	167	264,5	+ 35,6	15,7°	x
5	4	9	800	123	286,2	+ 43,9	7,7°	x
6	4	13	700	394	8,9	+ 42,9	16,7°	
7	5	10	1600	338	282,7	+ 30,1	12,9°	x
8	4	10	0	162	-	-	-	
9	4	10	100	42	318,7	+ 49,5	25,1°	x
10	4	10	400	175	44,1	+ 50,5	25,1°	
11	5	9	300	7,1	300,4	+ 34,7	12,2°	
mean: 5 sites					286,6	+ 44,5	16,9°	x

S: sites: 1-10 Cerviero; 11: Colle Vaccaro. N: number of samples. n: number of specimens. Oe: demagnetizing field applied. M: magnetic intensity. C: correction for dip applied. Mean direction includes sites 1, 4, 5, 7, 9 and α_{95} is calculated from site means; site 3 excluded because of large scatter; for exclusion of other sites see text.

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rez et al. 1974 in fig. 1, on the displacement of the Northern Calabrian crust.

The rotation of the Verbicaro unit can still be interpreted according to two different geological hypotheses:
i - the rotation is due to the movement of the Verbicaro nappe only, or that of a local pile of nappes, unrelated to the deep crust. In this case the paleomagnetic data represents tectonic movements within this area, which might be differential with respect of other tectonic units.
ii - no large differences of directions of motion or rotation about vertical axes occurred between the nappes. In this case the data represents the geotectonic motion of the North-Calabrian crustal block.

The metamorphic event suggests that the rocks were buried deep in the Calabrian crust and this fact, in my opinion, brings some evidence to the second hypothesis.

The differential rotation with respect of Sardinia shows that, if a Corsican-Sardinian microplate did exist, Northern Calabria, or at least its sedimentary cover, was detached from it during the Tertiary and was subject to indipendent motion.

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